

The Race between Man and Machine

Economic results, numerical checks, and a partial Lean formalization

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github.com/fgalarzac/ai-06-restrepo

The aggregate economy is the unit of analysis

Question. Can automation displace labor without making labor redundant?

A unit measure of tasks occupies $[N - 1, N]$. Capital can perform tasks below the automation frontier I . Labor has comparative advantage in higher-indexed tasks because $\gamma(i)$ is strictly increasing.

$$\underbrace{[N - 1, I^*)}_{\text{capital tasks}} \quad \underbrace{(I^*, N]}_{\text{labor tasks}}, \quad I^* = \min\{I, \tilde{I}\}.$$

Automation raises I . New tasks raise N . Their relative pace determines factor prices, employment, the labor share, and growth.

Source: AER Sections 1–4; NBER Working Paper 22252, Sections 1–4.

Firms choose the task allocation

For task i , firms compare the rental cost of capital with the effective labor cost:

$$p(i) = \min \left\{ R, \frac{W}{\gamma(i)} \right\}, \quad \frac{W}{R} = \gamma(\tilde{I}).$$

Threshold rule

For $i < I^*$, capital is technologically available and cheaper. For $i > I^*$, automation is unavailable or labor is cheaper.

In the full model, innovators allocate scientists between automation and new tasks:

$$S_I + S_N \leq S, \quad \dot{I} = \kappa_I S_I, \quad \dot{N} = \kappa_N S_N.$$

Source: NBER equations (3), (6), and (22).

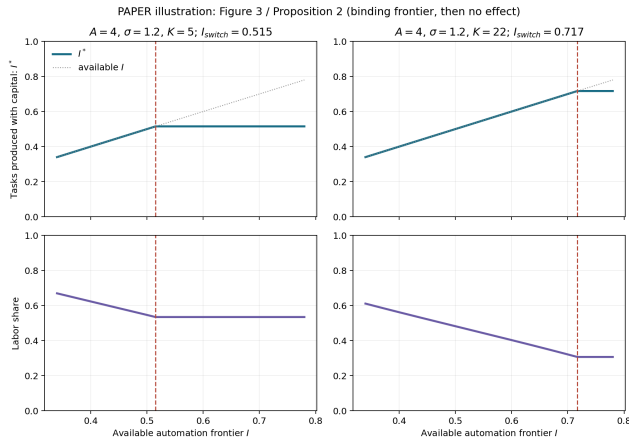
Automation affects four different outcomes

Outcome	Condition and conclusion
Fixed-capital wage	Productivity competes with displacement. The wage can rise or fall.
Long-run wage	On the active interior balanced growth path, capital adjustment can make the wage rise after a permanent automation lead.
Employment and labor share	A permanent automation lead lowers both, even when the long-run wage rises.
Welfare	With frictionless labor choice, automation raises welfare. A binding labor-supply constraint can reverse the result.

The paper's headline question therefore has separate short-run, long-run, distributional, and welfare answers.

Source: AER Propositions 3, 5, 6, and 9.

Numerical check: task allocation



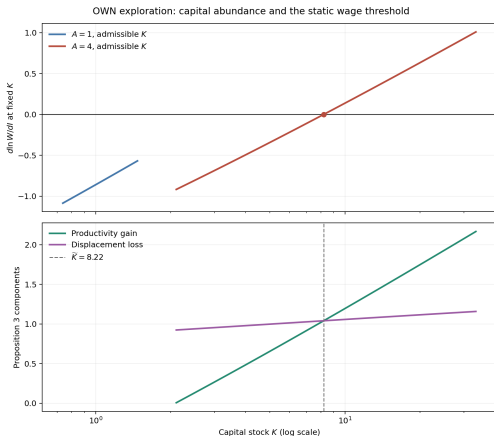
Source: Numerical illustration of Proposition 2 and Figure 3; generated by `sim.py`.

Paper result illustrated

The realized cutoff follows l while automation binds. Once the cost-equality threshold binds, extra available automation leaves the allocation unchanged.

The lower panels show the corresponding labor-share response.

Numerical check: the wage threshold



Own exploration

The fixed-capital wage derivative changes sign when the productivity gain overtakes the displacement term.

The analytical derivative agrees with a centered finite difference in the simulation.

Source: Original numerical exploration based on the static CES model; generated by `sim.py`.

Hand check of the AI-generated derivation

Appendix B supplies

$$s_L d \ln W + (1 - s_L) d \ln R = d \ln Y|_{K,L}, \quad (\text{B9})$$

$$d \ln W - d \ln R = q. \quad (\text{B10})$$

Adding $(1 - s_L)$ times (B10) to (B9) isolates the wage:

$$d \ln W = d \ln Y|_{K,L} + (1 - s_L)q.$$

Subtracting s_L times (B10) isolates the rental rate:

$$d \ln R = d \ln Y|_{K,L} - s_L q.$$

For automation, $q = -D$ with $D > 0$, so the wage equals productivity minus displacement.

Source: NBER Appendix B, equations (B9)–(B10).

Photo to insert

Place the handwritten check at `hand/derivation.jpg`. The slide will include it automatically.

Verdict on the AI output. The algebra is correct. It establishes the sign decomposition conditional on (B9) and (B10). It does not derive those identities from the continuum model.

Lean formalization architecture

1. **Source authority.** NBER Working Paper 22252, revised June 2017, pinned by SHA-256.
2. **Model declarations.** `Assumptions.lean` defines comparative advantage, productivity positivity, automation availability, and the cost threshold.
3. **Transparent targets.** `PaperInterface.lean` exposes two source-facing propositions as `Prop` values.
4. **Proofs.** `MainTheorems.lean` proves the cutoff and the Proposition 3 algebra. `ProofInterface.lean` supplies exact-type endpoints.
5. **Audit trail.** The source map records every named result and marks all unformalized results as deferred.

Current scope

The run is partially formalized. It proves two selected static results and records the remaining theory as proof debt.

Lean result 1: the static task threshold

Mathematical claim

$$I^* = \min\{I, \tilde{I}\}, \quad i < I^* \Rightarrow R < \frac{W}{\gamma(i)}.$$

Lean statement and proof step

```
theorem staticTaskThreshold_impl ... :  
  (forall task, task < EquilibriumTaskThreshold I threshold ->  
    AutomationAvailable I task /\ rental < wage / productivity task) /\ ... := by  
  have hProductivity : productivity task < productivity threshold :=  
    hComparativeAdvantage htaskThreshold  
  rw [hThreshold] at hProductivity  
  have hBeforeCleared : productivity task * rental < wage :=  
    (lt_div_iff0 hRentalPositive).mp hProductivity
```

Interpretation. Strict monotonicity orders task productivity. Equation (6) replaces $\gamma(\tilde{I})$ by W/R . Positivity lets Lean clear the denominators without reversing the inequality.

Source: `lean/MainTheorems.lean`, `theorem staticTaskThreshold_impl`.

Lean result 2: Proposition 3 algebra

1. Original equations

$$s_L w + (1 - s_L)r = p, \quad w - r = q.$$

2. Lean statement and proof fragment

```
theorem factorPriceDecomposition_impl ...  
  (hIncome : s*w + (1-s)*r = p)  
  (hRelative : w-r = q) :  
  w = p + (1-s)*q /\ r = p - s*q := by  
  constructor  
  · linear_combination hIncome + (1-s) * hRelative  
  · linear_combination hIncome - s * hRelative
```

3. What Lean verifies

- ▶ All variables live in $\mathbb{R}$ and both equations are explicit hypotheses.
- ▶ Each proof line records the exact elimination used in the hand derivation.
- ▶ Lean verifies the algebra. It does not yet derive (B9) or (B10) from the production model.

Source: `lean/MainTheorems.lean`; GitHub Actions run 35851458024.

Verified results and open boundary

Lean verifies

- ▶ The static task threshold and both cost comparisons.
- ▶ The factor-price formulas implied by (B9) and (B10).
- ▶ The automation sign split: rental rate positive; wage sign set by productivity versus displacement.
- ▶ Zero sorry, admit, or added axioms.

Still open

- ▶ Derivation of (B9) and (B10) from the continuum model.
- ▶ Exact CES coefficient positivity.
- ▶ Propositions 1–2 and the dynamic results.
- ▶ Independent semantic closeout for every named result.

Recorded check

```
lake build +AR18RaceManMachine: 8,317 jobs passed.  
paper_contribution.py check AR18RaceManMachine -fast: passed.
```

The NBER capital-threshold clause remains unresolved

The pinned NBER Proposition 3 prints

$$\bar{K} > \underline{K},$$

while Assumption 3 imposes $K < \underline{K}$. The same proposition assigns a negative wage effect to $K > \bar{K}$.

Inside Assumption 3, the stated negative-wage region cannot occur. The later AER version reverses the economic threshold, but the Lean run uses the NBER paper as its source authority.

Formalization decision

The run records this as AR18-P3-CAPITAL-THRESHOLD-01. It does not silently substitute the AER wording or claim a complete proof of Proposition 3.

Source: NBER page 13 and Assumption 3; AER Proposition 3 used only for comparison.

Conclusion

Automation does not have one uniform effect on labor.

- ▶ The fixed-capital wage reflects productivity minus displacement.
- ▶ Capital adjustment can reverse the long-run wage response while employment and labor share remain lower.
- ▶ Lean confirms the cutoff logic and the factor-price algebra, and exposes the assumptions that the current proof still accepts rather than derives.

Main lesson. Formal verification sharpens the economic claim only when the source equation, added assumptions, and proof boundary remain visible.

Source: Repository: github.com/fgalarzac/ai-06-restrepo.